



PENRITH HIGH SCHOOL

**2015
PRELIMINARY YEARLY EXAMINATION**

Mathematics

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Answer all Questions in the booklets provided

Total marks–70

SECTION I

Pages 3–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 6–9

60 marks

- Attempt Questions 11–14
- Allow about 1 hours 45 minutes for this section

Student Name: _____

Teacher Name: _____

This paper MUST NOT be removed from the examination room

Assessor: R Young

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the provided multiple-choice answer sheet for Questions 1–10

1. What is the value of $\frac{\pi^2}{6}$, correct to 3 significant figures?

- (A) 1.64
- (B) 1.65
- (C) 1.644
- (D) 1.645

2. What are the solutions of $2x^2 - 8x - 3 = 0$?

- (A) $x = \frac{-8 \pm \sqrt{40}}{4}$
- (B) $x = \frac{8 \pm \sqrt{40}}{4}$
- (C) $x = \frac{7 \pm \sqrt{88}}{4}$
- (D) $x = \frac{-7 \pm \sqrt{88}}{4}$

3. $x^3 - 27$ in factorised form is

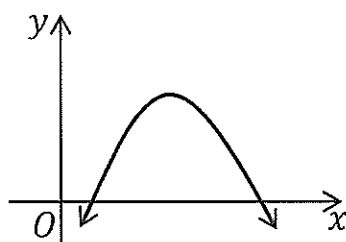
- (A) $(x + 3)(x^2 + 3x + 9)$
- (B) $(x - 3)(x^2 + 3x + 9)$
- (C) $(x - 3)(x^2 + 6x + 9)$
- (D) $(x + 3)(x^2 + 6x + 9)$

4. Which of the following is true?

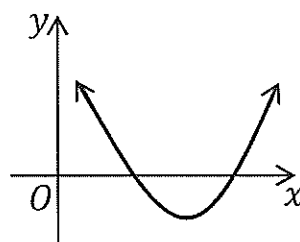
- (A) An even function has a point of symmetry at the origin.
- (B) An even function is symmetrical about the x – axis.
- (C) An odd function has point symmetry about the origin.
- (D) An odd function is symmetrical about the y – axis.

5. Which graph represents a positive definite function.

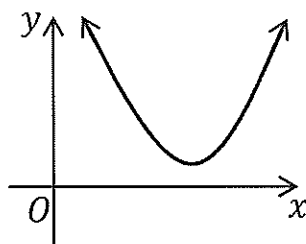
(A)



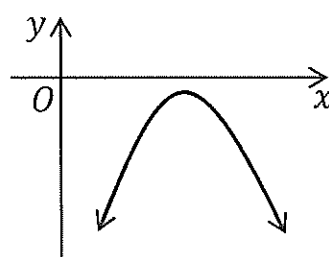
(B)



(C)



(D)



6. Which statement defines the domain of the function $f(x) = \sqrt{4 - x^2}$?

- (A) $x < 4$
- (B) $-4 \leq x \leq 4$
- (C) $-2 \leq x \leq 2$
- (D) $-2 < x < 2$

7. The exact value of $\cos(-135^\circ)$ is:

- (A) $\frac{1}{\sqrt{2}}$
- (B) $\frac{-1}{\sqrt{2}}$
- (C) -0.707
- (D) 0.707

8. If $\sec\theta = \frac{-5}{4}$ and $\tan\theta > 0$ then,
- (A) $\sin\theta = \frac{3}{5}$
- (B) $\sin\theta = -\frac{3}{5}$
- (C) $\cos\theta = \frac{4}{5}$
- (D) $\tan\theta = -\frac{3}{5}$
9. The equation of the normal to the curve $y = x^2 + 2x$ at the point where $x = 2$
- (A) is parallel to the x – axis.
- (B) is perpendicular to the x – axis.
- (C) cuts the x – axis at a positive value of x .
- (D) cuts the x – axis at a negative value of x .
10. The derivative of $f(x) = \frac{1}{x}$ at $x = 0$,
- (A) is positive
- (B) is negative
- (C) $\pm\infty$
- (D) does not exist

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

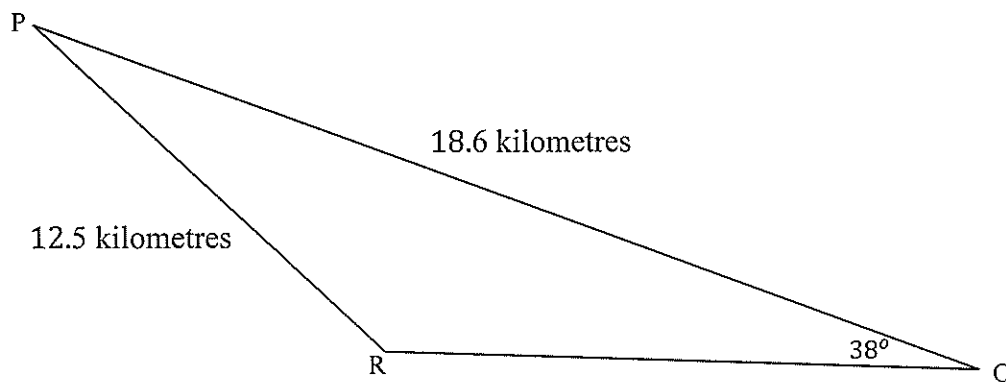
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 11–14, your responses should include relevant mathematical reasoning and/or calculations

Question 11 (15 marks) Use a new writing booklet.

- a) Rationalise the denominator of $\frac{1}{\sqrt{5}-3}$ 2
- b) Factorise $2x^2 + 5x - 3$ 2
- c) Solve for x , $|3x + 2| \leq 3$ 2
- d) Express $\frac{2x}{x^2+2x-3} + \frac{2}{x-1}$ as a single fraction in simplest form. 2
- e) Shade the region that satisfies $y > 3x - 1$ and $y \leq 5$ simultaneously. 3
- f) Prove that $\tan\theta + \cot\theta = \operatorname{cosec}\theta \sec\theta$. 2

g)



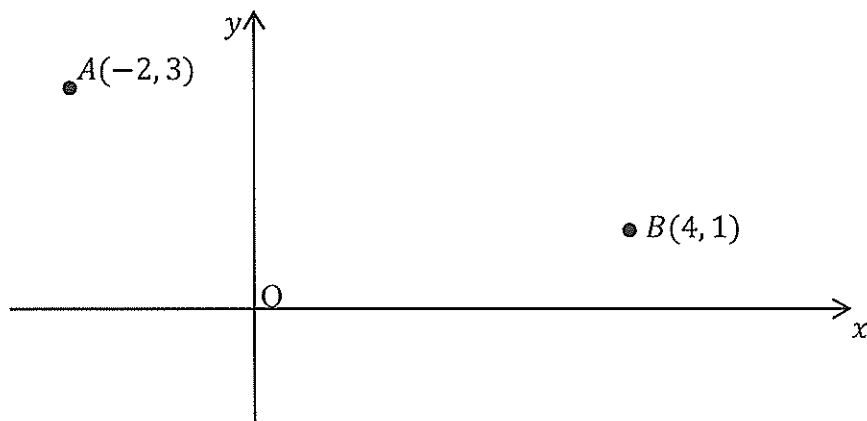
In the diagram above, $PQ = 18.6$ kilometres, $PR = 12.5$ kilometres and $\angle PQR = 38^\circ$. $\angle PRQ$ is obtuse. Find the size of $\angle PRQ$ correct to the nearest minute.

2

Question 12 (15 marks) Use a new writing booklet.

In the diagram above, A, B and O are the points $(-2, 3)$, $(4, 1)$ and $(0, 0)$ respectively.

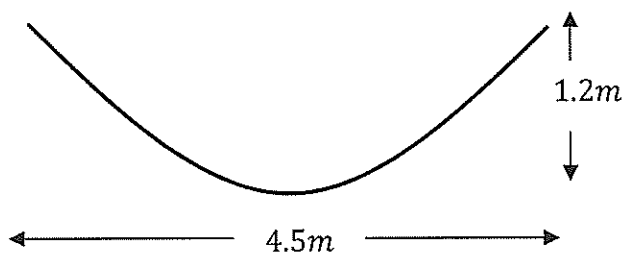
a)



- | | | |
|--|---|---|
| i. | Show that the equation of AB is $x + 3y - 7 = 0$ | 1 |
| ii. | Find the distance AB | 1 |
| iii. | Find M, the midpoint of AB | 1 |
| iv. | Show that the equation of the line through M, perpendicular to AB is $3x - y - 1 = 0$. | 1 |
| v. | Hence find D the point of intersection with the y axis of the line in part iv. | 1 |
| vi. | Hence find the area of the triangle BAD. | 2 |
| b) If α and β are the roots of the equation $2x^2 - 3x + 7 = 0$, find: | | |
| i. | $\alpha + \beta$ and $\alpha\beta$ | 1 |
| ii. | $\alpha\beta - \alpha - \beta + 1$ | 1 |
| c) For what values of k does the equation $4x^2 - 4x + k = 0$ have real roots? | | 2 |
| d) Solve $3^{2x} - 4 \cdot 3^x + 3 = 0$ | | 2 |
| e) If one root of $3x^2 - 8x + k = 0$ is three times the other root, find k . | | 2 |

Question 13 (15 marks) Use a new writing booklet.

- a) A parabola is defined by $x^2 = 12y$.
- i. What is the coordinates of the focus? 1
- ii. What is the equation of the directrix? 1
- b) Find the centre and radius of the circle $x^2 - 4x + y^2 - 2y - 20 = 0$. 2
- c) A point $P(x, y)$ moves so that AP is perpendicular to PB . Given $A = (2, 1)$ and $B = (-3, 6)$, find the of the locus of P . 3
- d) Find the equation of the parabola with vertex $(2, 4)$ and focus $(-3, 4)$. 2
- e) Find the equation of the locus of a point that is always 3 units from the line $3x + 4y - 1 = 0$. 2
- f) A satellite dish is to be 4.5 m wide and 1.2 m deep. How far from the vertex is the focus? 2



- g. Find the equation of the locus of the midpoints of all chords of length 4 units of the circle with equation $x^2 + y^2 - 4x + 2y = 4$. 2

Question 14 (15 marks) Use a new writing booklet.

a) Differentiate the following with respect to x .

i. $2x^2 - 7x + 1$ 1

ii. $\frac{x^2 + 3x + 6}{x}$ 2

iii. $\frac{5}{\sqrt[3]{x}}$ 2

b) Differentiate $f(x) = x^2 + 4x$ from first principles. 2

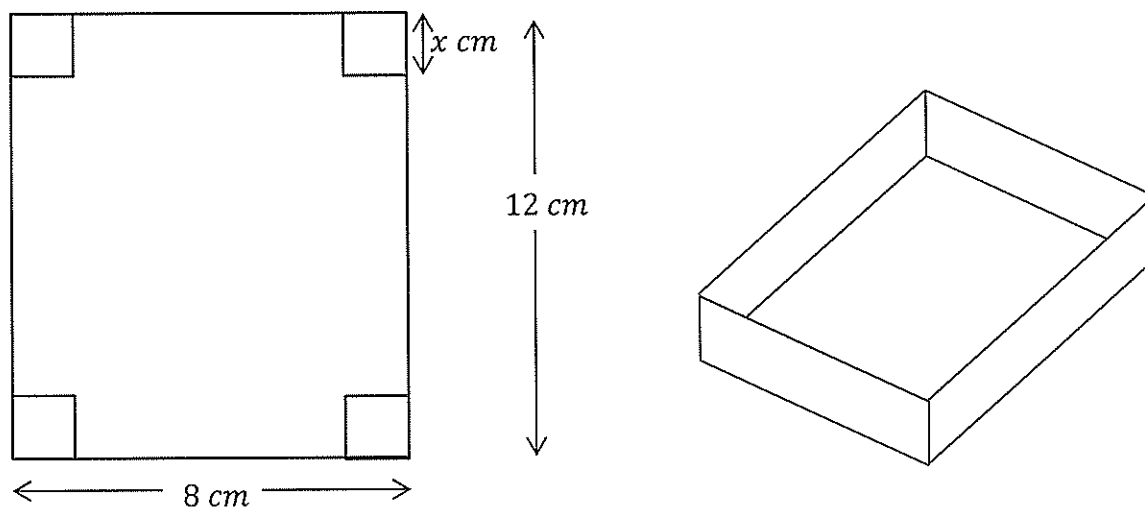
c) Find the value of $\lim_{x \rightarrow \infty} \frac{3x^2}{7x^2 + 1}$ 2

d) A function is defined by $f(x) = x^3 - 3x^2 - 9x + 22$.

i. Find the coordinates of the stationary points and determine their nature. 2

ii. Sketch the graph of $y = f(x)$ showing the turning points and where the curve meets the y axis. 1

e) An open rectangular prism is to be made from a flat sheet of metal $12\text{cm} \times 8\text{cm}$. Squares of side $x\text{cm}$ are cut from each corner then the metal is bent to form the prism.



i. Show that the volume of the prism is given by $V = 4x^3 - 40x^2 + 96x$. 1

ii. Find the value of x that maximises the volume of the prism. Leave your answer in exact form. 2

End of paper

YEAR 11 YEARLY MATHEMATICS SOLUTIONS

Multiple Choice

1.A 2.B 3.B 4.C 5.C 6.C 7.B 8.B 9.C 10.D

Question 11

$$a. \frac{1}{\sqrt{5}-3} \times \frac{\sqrt{5}+3}{\sqrt{5}+3}$$

$$= \frac{\sqrt{5}+3}{5-9}$$

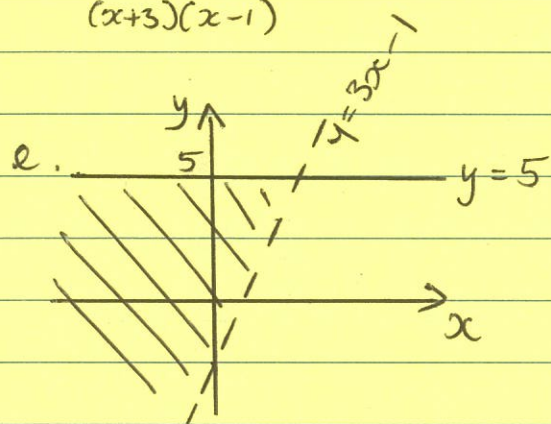
$$= \frac{\sqrt{5}+3}{-4} \text{ or } -\frac{5-3}{4}$$

$$b. (2x-1)(x+3)$$

$$c. \begin{array}{ll} 3x+2 \leq 3 & 3x+2 \geq -3 \\ 3x \leq 1 & 3x \geq -5 \\ x \leq \frac{1}{3} & x \geq -\frac{5}{3} \end{array}$$

$$d. \frac{2x}{(x+3)(x-1)} + \frac{2(x+3)}{(x-1)(x+3)}$$

$$= \frac{4x+6}{(x+3)(x-1)}$$



$$f. \text{LHS} = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta} = \operatorname{cosec} \theta \sec \theta = \text{RHS}$$

$$g. \frac{\sin \angle PRQ}{18.6} = \frac{\sin 38^\circ}{12.5}$$

$$\sin \angle PRQ = \frac{18.6 \sin 38^\circ}{12.5}$$

$$\sin \angle PRQ = 0.9161 \dots$$

$$\angle PRQ = 66.36^\circ \text{ or } 113.63^\circ$$

$$\text{Obtuse } \angle PRQ = 113.63^\circ$$

Question 12

$$ai. m = \frac{3-1}{-2-4} = -\frac{1}{3}$$

$$y-1 = -\frac{1}{3}(x-4)$$

$$3y-3 = -x+4$$

$$x+3y-7=0$$

$$ii. AB = \sqrt{(1-3)^2 + (4+2)^2} \\ = \sqrt{4+36} \\ = \sqrt{40} = 2\sqrt{10}$$

$$iii. M: x = \frac{-2+4}{2} = 1$$

$$y = \frac{3+1}{2} = 2 \quad M(1, 2)$$

$$iv. m_{\perp} = 3 \quad y-2 = 3(x-1)$$

$$y-2 = 3x-3$$

$$3x-y-1=0$$

Question 12 continued

$$\text{av. } x=0 \quad 0-y-1=0$$

$$y=-1$$

$$\therefore D(0, -1)$$

$$\text{vi. distance} = \sqrt{(2+1)^2 + (1-0)^2}$$

$$= \sqrt{10}$$

$$\text{Area} = \frac{1}{2} \times 2\sqrt{10} \times \sqrt{10}$$

$$= 10 \text{ units}^2$$

$$\text{bi. } \alpha + \beta = \frac{3}{2} \quad \alpha\beta = \frac{7}{2}$$

$$\text{ii. } \alpha\beta - (\alpha + \beta) + 1$$

$$= \frac{7}{2} - \frac{3}{2} + 1$$

$$= 3$$

$$\text{c. } \Delta \geq 0$$

$$\Delta = b^2 - 4ac$$

$$4^2 - 4 \cdot 4 \cdot k \geq 0$$

$$16(1-k) \geq 0$$

$$1 \geq k$$

$$k \leq 1$$

$$\text{d. } (3^x)^2 - 4(3^x) + 3 = 0$$

$$\text{let } m = 3^x$$

$$m^2 - 4m + 3 = 0$$

$$(m-3)(m-1) = 0$$

$$m = 3, 1$$

$$3^x = 3 \quad 3^x = 1$$

$$x = 1, 0$$

$$\text{e. roots } \alpha \text{ and } 3\alpha$$

$$4\alpha = \frac{8}{3} \quad 3\alpha^2 = \frac{k}{3}$$

$$\alpha = \frac{2}{3}$$

$$3\left(\frac{2}{3}\right)^2 = \frac{k}{3}$$

$$k = 4$$

Question 13

$$\text{ai. } x^2 = 12y \quad x^2 = 4ay$$

$$= 4 \times 3y$$

$$\text{focus } (0, 3)$$

$$\text{ii. } y = -3.$$

$$\text{b. } x^2 - 4x + 4 + y^2 - 2y + 1 = 25$$

$$(x-2)^2 + (y-1)^2 = 25$$

$$\text{centre } (2, 1) \text{ radius} = 5$$

$$\text{c. } m_{PA} = \frac{y-1}{x-2} \quad m_{PB} = \frac{y-5}{x+3}$$

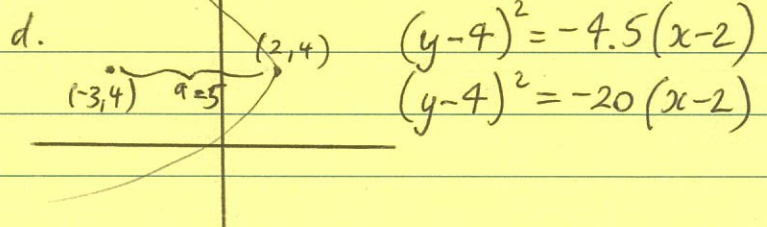
$$m_{PA} \times m_{PB} = -1$$

$$\frac{y-1}{x-2} \times \frac{y-5}{x+3} = -1$$

$$y^2 - 6y + 5 = -(x^2 + x - 6)$$

$$y^2 - 6y + 5 = -x^2 - x + 6$$

$$x^2 + x + y^2 - 6y - 1 = 0$$



$$\text{e. } PA = 2PB$$

$$\sqrt{(x+2)^2 + y^2} = 2\sqrt{(x-1)^2 + y^2}$$

$$x^2 + 4x + 4 + y^2 = 4[x^2 - 2x + 1 + y^2]$$

$$0 = 3x^2 - 12x + 3y^2$$

$$0 = x^2 - 4x + y^2$$

Question 13 continued

f. $x^2 = 4ay$ (2, 4)

$$(2)^2 = 4a(2)$$

$$4 = 8a$$

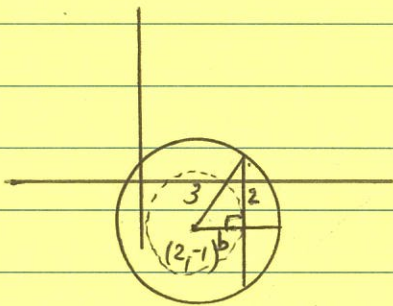
$$a = \frac{1}{2}$$

g. $x^2 - 4x + y^2 + 2y = 4$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = 9$$

$$(x-2)^2 + (y+1)^2 = 9$$

centre (2, -1) radius = 3



$$b^2 = 3^2 - 2^2$$

$$= 9 - 4$$

$$b = \sqrt{5}$$

$\therefore$ locus is circle with centre (2, -1) and radius $\sqrt{5}$

$$(x-2)^2 + (y+1)^2 = 5$$

Question 14

a.i. $4x - 7$

ii. $y = x + 3 + 6x^{-1}$

$$y' = 1 - 6x^{-2}$$

iii. $y = 5x^{-\frac{1}{3}}$
 $y' = -\frac{5}{3}x^{-\frac{4}{3}}$

b. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 4(x+h) - (x^2 + 4x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - x^2 - 4x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 4h}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 4$$

$$= 2x + 4$$

c. $\lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2}}{\frac{7x^2}{x^2} + \frac{1}{x^2}}$

$$= \lim_{x \rightarrow \infty} \frac{3}{7 + \frac{1}{x^2}}$$

$$= \frac{3}{7 + \frac{1}{\infty^2}} = \frac{3}{7}$$

d.i. $f(x) = x^3 - 3x^2 - 9x + 22$

$$f'(x) = 3x^2 - 6x - 9$$

$$0 = 3(x^2 - 2x - 3)$$

$$0 = (x-3)(x+1)$$

$$x = 3, -1$$

$$f(3) = -5 \quad f(-1) = 27$$

$$(3, -5) \quad (-1, 27)$$

x	-2	-1	0
f(x)	15	0	-9

↗ ↖ ↘ ↙

(-1, 27) max

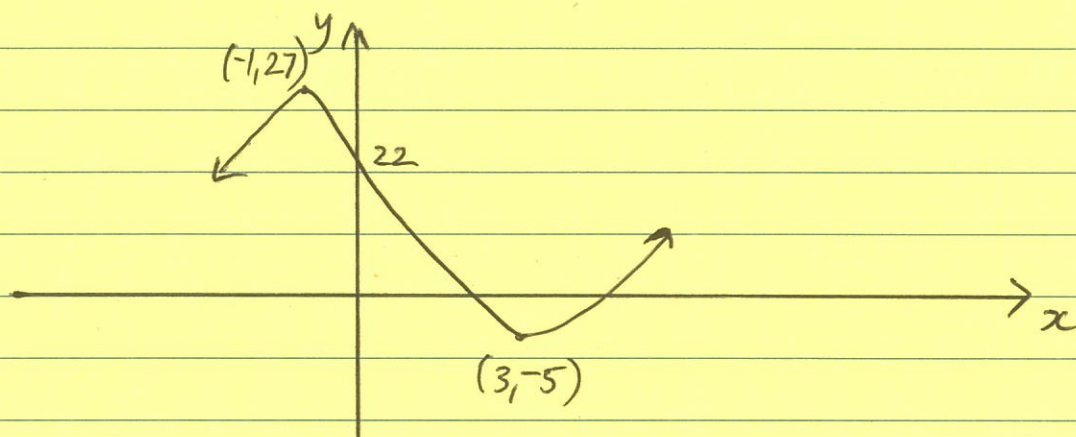
x	2	3	4
f(x)	-9	0	15

↘ ↗ ↙ ↘

(3, -5) min

Question 14 continued

dii. $f(x) = x^3 - 3x^2 - 9x + 22$



e.i. $V = x(8 - 2x)(12 - 2x)$
 $= (8x - 2x^2)(12 - 2x)$
 $= 96x - 40x^2 + 4x^3$

ii. $\frac{dV}{dx} = 12x^2 - 80x + 96$
 $4(3x^2 - 20x + 24) = 0$

$$x = \frac{20 \pm \sqrt{20^2 - 4 \cdot 3 \cdot 24}}{6}$$

$$= \frac{20 \pm \sqrt{112}}{6}$$

$$= \frac{10 \pm 2\sqrt{7}}{3}$$

test	x	5	$\frac{10+2\sqrt{7}}{3}$	6	x	1	$\frac{10-2\sqrt{7}}{3}$	2
	$\frac{dV}{dx}$	-1	0	12		7	0	-4
		$\nwarrow$	$\longleftrightarrow$	$\nearrow$		$\nwarrow$	$\longleftrightarrow$	$\nwarrow$

$\therefore$ max volume when $x = \frac{10 - 2\sqrt{7}}{3}$